

An Axial-Flux Flywheel Generator, Taken Apart on Paper

The physics of a machine I built out of PVC pipe, copper plumbing tube and 3D-printed plastic at fifteen — worked out properly, fifteen years of schooling later.

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What this is. In 2021, locked down and fifteen, I built a generator: two 3D-printed rotor disks carrying eight permanent magnets each, spinning either side of four hand-wound coils, on a copper-pipe shaft in a PVC frame. It lit LEDs. A gym weight plate bolted to the shaft served as a flywheel; a hand crank through a 4:1 planetary gearbox drove it; a 200 g mass on a string over a spool gave it a repeatable, calculable energy input.

I did not have this mathematics then. I had a YouTube video about flywheel batteries, a soldering iron and a first prototype that destroyed itself. This document is the analysis I could not write at the time: what the machine actually does, which numbers govern it, why the first version collapsed, and what the measurements caught on video really say. Where a quantity was measured it is marked **MEASURED**; where it is estimated from photographs or standard part sizes it is marked **EST.** and the estimate is stated. Nothing here is tuned to make the machine look better than it was.

1 The machine

The topology is a **dual-rotor, coreless axial-flux permanent-magnet generator**. Two disks of magnets rotate on a common shaft; a stationary disk of air-cored coils sits in the gap between them. Flux crosses the gap parallel to the shaft — axially — so every magnet passes squarely over a coil once per pole. Almost every small wind-turbine generator built by hobbyists uses this arrangement, for the same reason I ended up at it: it needs no iron core, no laminations, no machining, and it tolerates a builder's tolerances.

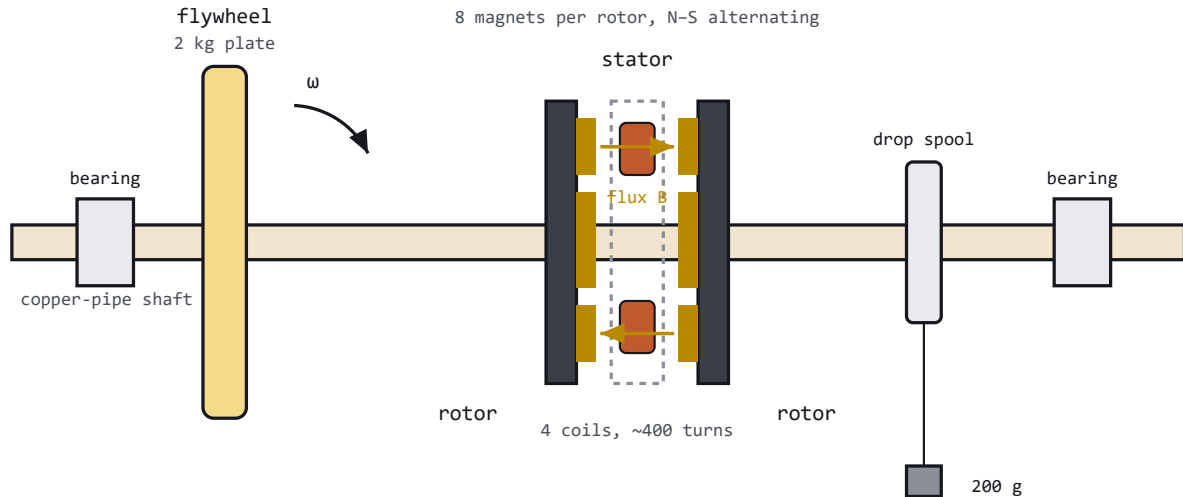


Figure 1 — Section through the machine. Flux crosses the air gap axially, from one rotor disk to the other, passing through the stationary coils on the way. The flywheel plate and the drop-spool sit on the same shaft; the hand crank and gearbox (not shown) drive the left-hand end. Drawn to describe the layout, not to scale.

Three consequences follow from having no iron in the stator, and all three mattered to a fifteen-year-old with a hot-glue gun:

- **No cogging.** With an iron-toothed stator, the magnets snap from tooth to tooth and the shaft resists being turned even with the output open-circuit. A coreless stator has no preferred angular position, so the machine spins freely under no load — which is exactly what you want when the input is a falling 200 g mass with 2 J to its name.
- **No core loss.** There is no iron to carry hysteresis or eddy-current losses, so at these speeds essentially all the electrical loss is I^2R in the copper.
- **A large effective air gap, and a weak field to show for it.** The flux has to cross the coil thickness plus two mechanical clearances instead of a millimetre of air. This is the price of the other two, and §2 shows how heavily it is paid.

2 Where the voltage comes from

Faraday's law is the whole machine in one line. The flux threading one turn of a coil is the field integrated over the turn's area,

$$\Phi = \int_A \mathbf{B} \cdot d\mathbf{A} \quad (1)$$

and the electromotive force appearing across a coil of N turns is the rate at which that flux changes, with the minus sign of Lenz's law recording that the induced current opposes the change that made it:

$$\varepsilon = -N \frac{d\Phi}{dt} \quad (2)$$

Nothing in the coil knows about rotation; it only knows that the flux through it is changing. As a magnet sweeps across a coil the flux rises from zero to a maximum and back; as the next magnet — reversed in polarity — arrives, it swings equally far negative. To a good approximation for a sinusoidally-varying overlap,

$$\Phi(t) \approx \hat{\Phi} \sin(\omega_e t) \Rightarrow \varepsilon(t) = -N\hat{\Phi} \omega_e \cos(\omega_e t) \quad (3)$$

where ω_e is the *electrical* angular frequency. It is not the shaft speed. Each pair of magnets — one north-facing, one south-facing — gives one full electrical cycle per pass, so with p pole pairs on the rotor,

$$\omega_e = p \omega_m, \quad f = \frac{p n}{60} \quad [\text{Hz, with } n \text{ in rpm}] \quad (4)$$

My rotors carry eight magnets each, alternating in polarity, so $p = 4$. At 300 rpm the coils see 20 Hz — mains-like frequencies out of a hand crank, purely because the pole count does the work that speed would otherwise have to. Putting (3) and (4) together gives the peak EMF of the whole stator, with N_{tot} turns in series:

$$\hat{\varepsilon} = N_{\text{tot}} \hat{\Phi} p \omega_m \equiv k \omega_m \quad (5)$$

THE ONE EQUATION TO KEEP

Voltage is proportional to speed, and the constant of proportionality $k = N_{\text{tot}} \hat{\Phi} p$ is fixed by construction: turns, flux per pole, pole pairs. There is no other knob. Every design decision in this project — more turns, thinner wire, smaller air gap, more magnets, a gearbox to raise ω_m — is an attempt to move one of those four terms.

What the geometry predicts

$\hat{\Phi}$ is set by the field and by the area the turns actually enclose — and it is the *smaller* of the two apertures that governs. The neodymium magnets present a 20×10 mm face, 200 mm^2 , but a turn wound on the 6 mm former with a 6 mm build has a mean diameter of 12 mm and encloses only

$$A_{\text{turn}} = \pi(D_{\text{mean}}/2)^2 = \pi(6 \times 10^{-3})^2 \approx 1.13 \times 10^{-4} \text{ m}^2 \quad (6a)$$

Crediting the turns with the magnet's area rather than their own is the easiest way to flatter a machine like this on paper, and it overstates the answer by nearly a factor of two. With $B_{\text{gap}} \approx 0.15$ T across the working gap — a fair figure for neodymium blocks facing each other across a coil a few millimetres thick — and the 400 turns per coil that §3 derives from the winding window, four coils in series give

$$k_{\text{ideal}} = 1600 \times (0.15 \times 1.13 \times 10^{-4}) \times 4 \approx 0.11 \text{ V}\cdot\text{s}/\text{rad} \quad (6b)$$

That predicts 4 V on the terminals at about 36 rad/s — 350 rpm. Hold on to the figure 0.11: §8 measures the same constant from an experiment that knows nothing about any of this geometry, and the comparison is the most instructive number in the document.

3 The coil as a real source

A coil is not an ideal voltage source; it is an EMF behind its own resistance. For enamelled copper of length ℓ and cross-section A_w ,

$$R = \frac{\rho \ell}{A_w}, \quad \rho_{\text{Cu}} = 1.68 \times 10^{-8} \Omega \cdot \text{m} = 0.0172 \Omega \cdot \text{mm}^2/\text{m} \quad (7)$$

AWG 32 wire is 0.203 mm across the bare copper — $A_w = 0.0324 \text{ mm}^2$, or 0.52 Ω per metre — and 0.25 mm measured over the enamel **MEASURED**, which is the figure that decides how many turns will physically fit.

How many turns actually went on

The winding window in the printed holder is 5 mm deep by 6 mm wide **MEASURED** — 30 mm^2 of cross-section to fill with 0.25 mm wire. Perfect square packing would put 16 turns in every square millimetre ($1/d_{\text{ins}}^2$); hand-winding without a jig or a counter realistically reaches $n_A \approx 12$ or 13:

$$N \approx A_{\text{window}} \cdot n_A \approx 30 \times 13 \approx 400 \text{ turns} \quad (\text{ceiling: 480 square-packed, 555 hexagonal}) \quad (8a)$$

This matters because I remembered these coils as thousand-turn coils for years, and the window simply will not hold a thousand: 555 is a hard geometric ceiling and hand-winding does not come close to it. The first-generation coils — the fat hand-wound lumps in the earliest photographs — were much stubbier and did carry more wire; the printed holder that replaced them took a sleeker winding, and it is the sleeker one whose numbers appear throughout this document.

Four hundred turns wound on the 6 mm former, building out 6 mm, gives a mean turn diameter of 12 mm and so

$$\ell = N \pi D_{\text{mean}} = 400 \times \pi \times 0.012 \approx 15 \text{ m} \Rightarrow R_{\text{coil}} \approx 7.8 \Omega \quad (8b)$$

Four in series is about 31 Ω of source resistance and some 18 g of copper — and 31 Ω is a number worth remembering when §8 gets to where the energy went. It is also the easiest prediction in this document to falsify: put an ohmmeter across one coil, and if it does not read somewhere near 8 Ω , equation (8a) is wrong and everything downstream of it moves.

Why more turns is not simply better

Adding turns raises the EMF linearly — but it also lengthens the wire, so the resistance rises linearly too. Into a load R_L the delivered power is

$$P_L = \frac{\varepsilon^2 R_L}{(R_{\text{coil}} + R_L)^2}, \quad \text{maximal at } R_L = R_{\text{coil}} \quad (9)$$

With $\varepsilon \propto N$ and $R_{\text{coil}} \propto N$, the short-circuit current $\varepsilon/R_{\text{coil}}$ is *independent of the number of turns* — it depends only on the winding window, the wire's resistivity and the rate of change of flux. Turns buy volts, never watts. What buys watts is copper cross-section in the window, flux, and speed.

WHAT I ACTUALLY NEEDED

The load was LEDs, which need *voltage* before they need current: below their forward drop they draw nothing at all. Winding many turns of thin wire was, by accident, exactly the right choice for the job — a high-voltage, high-impedance, low-current source. Had I been charging a battery instead, the same coils would have been close to useless, and the fix would have been fewer turns of much thicker wire.

4 Rectification, and the voltage floor for light

What comes out of the coils is alternating: each pole reverses the flux, so the terminals swing symmetrically about zero. An LED conducts one way only, and would flash at half duty and sit dark for the other half. Learning what a full-bridge rectifier was, and building one on a breadboard, was the first piece of electronics I did on this project.

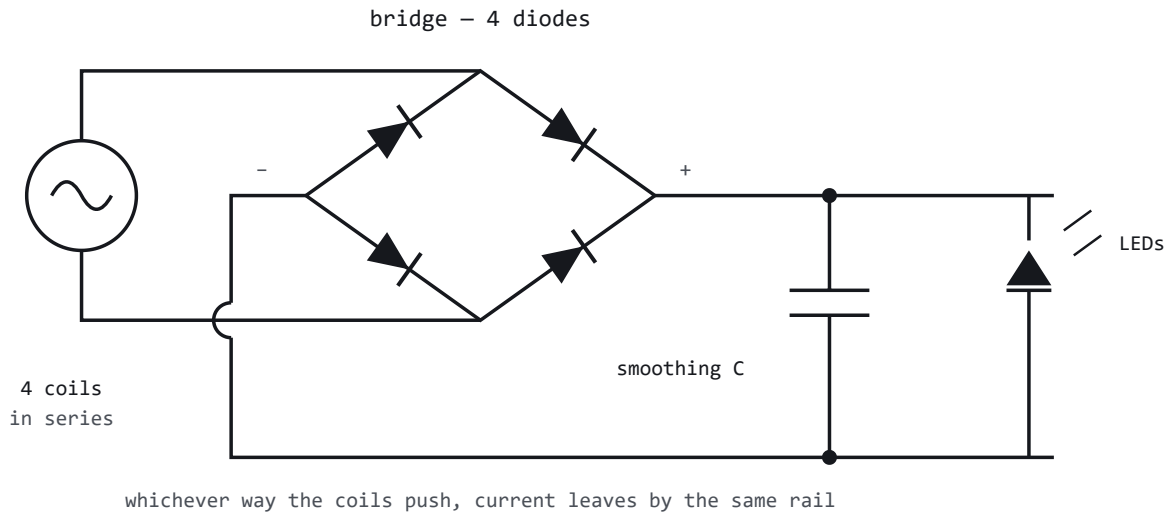


Figure 2 — The output stage as built. Four coils in series feed a full-bridge rectifier, a smoothing capacitor and the LEDs. Two diodes conduct on each half-cycle, so the bridge costs two forward drops, not one.

The bridge buys both half-cycles at the price of two silicon drops in series, and the LED then takes its own. For light to appear at all, the instantaneous EMF must clear

$$\hat{\epsilon} \geq V_{f,LED} + 2 V_{f,diode} \approx 3.0 + 2(0.7) = 4.4 \text{ V} \quad (10)$$

for blue LEDs, which are the expensive ones in forward drop — a red LED at 1.8 V would have lit at 3.2 V and made the machine look twice as good. Combining (10) with (5) gives the speed below which the machine is, visibly, a paperweight:

$$\omega_{\min} = \frac{V_{f,LED} + 2 V_{f,diode}}{k} \quad (11)$$

This threshold is not a nuisance; it is the design specification. Everything downstream — the gearbox ratio, the flywheel, how high I had to lift the 200 g mass — exists to keep ω_m above $\omega_{\min}$ for long enough to see something.

5 What the electrical load feels like at the shaft

Close the circuit and the coils carry current; that current sits in the magnets' field and produces a force opposing the motion that created it. This is Lenz's law arriving as a mechanical sensation — the crank gets heavier the moment the LEDs light. Quantitatively, with the same constant k from (5):

$$i = \frac{k\omega_m}{R_{\text{coil}} + R_L}, \quad \tau_{\text{brake}} = k i = \frac{k^2}{R_{\text{coil}} + R_L} \omega_m \quad (12)$$

The braking torque is proportional to speed: electrically, the machine behaves at the shaft exactly like a viscous damper with coefficient $k^2/(R_{\text{coil}}+R_L)$. The same k that generates volts generates drag, and its square governs how hard the machine fights back. Two things follow that took me a long time to understand as a teenager:

- An open-circuit generator is nearly free to spin, and a short-circuited one is nearly impossible to turn. Nothing mechanical changed.
- Power out and drag are the same phenomenon: $P = \tau_{\text{brake}}\omega_m = k^2\omega_m^2/(R_{\text{coil}}+R_L)$. There is no way to take energy out without being felt.

Equation (12) also explains the shape of the gravity-drop measurement in §8. A hanging mass applies a constant torque $m g r_s$ at the spool. Against a speed-proportional brake, the shaft accelerates toward a terminal speed $\omega_\infty = m g r_s(R_{\text{coil}}+R_L)/k^2$; below it, and with the machine's rotational inertia dominating, the speed ramps very nearly linearly — which is exactly what the voltmeter recorded.

6 The flywheel, and an honest reckoning of what it stores

The project began with a video about flywheel batteries, so a flywheel there had to be. A 2 kg cast gym plate fit the shaft adapter perfectly; the bore of a standard plate is a happy accident of engineering standardisation, and I tested up to 5 kg on the same shaft. For a plate of mass m , outer radius r_o and bore radius r_i :

$$I = \frac{1}{2} m(r_o^2 + r_i^2), \quad E = \frac{1}{2} I \omega_m^2 \quad (13)$$

With $r_o = 90$ mm and $r_i = 12.5$ mm, the 2 kg plate gives $I \approx 8.3 \times 10^{-3} \text{ kg}\cdot\text{m}^2$. Then:

SHAFT SPEED	Ω_m	STORED ENERGY	EQUIVALENT
100 rpm	10.5 rad/s	0.45 J	lifting the 200 g mass 23 cm
300 rpm	31.4 rad/s	4.1 J	lifting it 2.1 m
600 rpm	62.8 rad/s	16.3 J	lifting it 8.3 m
1000 rpm	104.7 rad/s	45.3 J	lifting it 23 m

Energy goes as the *square* of speed but only linearly with mass, which is why real flywheel storage is built out of carbon fibre spinning at tens of thousands of rpm rather than out of heavier steel. Doubling my plate to 4 kg would have doubled the stored energy; doubling the speed would have quadrupled it — and the machine would have run into a very different limit, since the burst hazard of a spinning disk scales with rim speed, not mass.

THE REALITY CHECK I COULD NOT DO AT FIFTEEN

At 300 rpm the flywheel holds about 4 joules. A single AA alkaline cell holds roughly 10 000 joules — two and a half *thousand* times more, at a fraction of the mass and with no bearings to keep it. My machine was never going to be a battery. What it was, and what it turned out to be worth, was a device that made an abstract chain — potential energy → kinetic energy → magnetic flux change → EMF → light — visible and measurable end to end, one link at a time.

7 Gearing: buying speed with torque

Since $\varepsilon \propto \omega_m$ and a comfortable hand crank runs at perhaps 60–90 rpm, the cheapest volts available were geometric. I designed a 4:1 planetary gearbox in CAD and printed it. In a planetary train with a fixed ring, sun

input and carrier output, the reduction is

$$G = 1 + \frac{Z_{\text{ring}}}{Z_{\text{sun}}} \quad (14)$$

so a ring with three times the teeth of the sun gives 4:1. Driven the other way — crank on the carrier, generator on the sun — the same train becomes a $4\times$ *speed increaser*, which is how I used it. The bargain is strictly conserved:

$$\omega_{\text{out}} = G \omega_{\text{in}}, \quad \tau_{\text{out}} = \frac{\eta \tau_{\text{in}}}{G}, \quad P_{\text{out}} = \eta P_{\text{in}} \quad (15)$$

Four times the voltage, one quarter of the torque available to make current with, and whatever the printed plastic teeth lose to friction on top. Combining with (12), the torque I had to supply at the crank against the electrical load goes as G^2 : the gearbox raises ω at the generator, which raises both the induced current and the arm through which it acts. The crank got noticeably heavier once the ratio went in, and that heaviness was, correctly, the feeling of delivering more power.

8 The gravity-drop experiment

The last thing I added was the best idea in the project, though I did not think of it that way at the time. A hand crank is not a measurement: nobody can turn a handle at a repeatable speed, and there is no way to say how much energy you put in. So I printed a spool, wound a string on it, hung a 200 g mass from it and dropped it from 1 m. That is a known, repeatable energy input, delivered as a constant torque:

$$E_{\text{in}} = m g h = 0.200 \times 9.81 \times 1.00 = 1.96 \text{ J}, \quad \tau_{\text{in}} = m g r_s, \quad P_{\text{in}}(t) = m g v(t) \quad (16)$$

What the video actually recorded

The multimeter is visible in frame throughout the drop, on its 20 V range, reading the rectified output. Stepping the footage frame by frame gives a genuine time series — the only quantitative data this project ever produced:

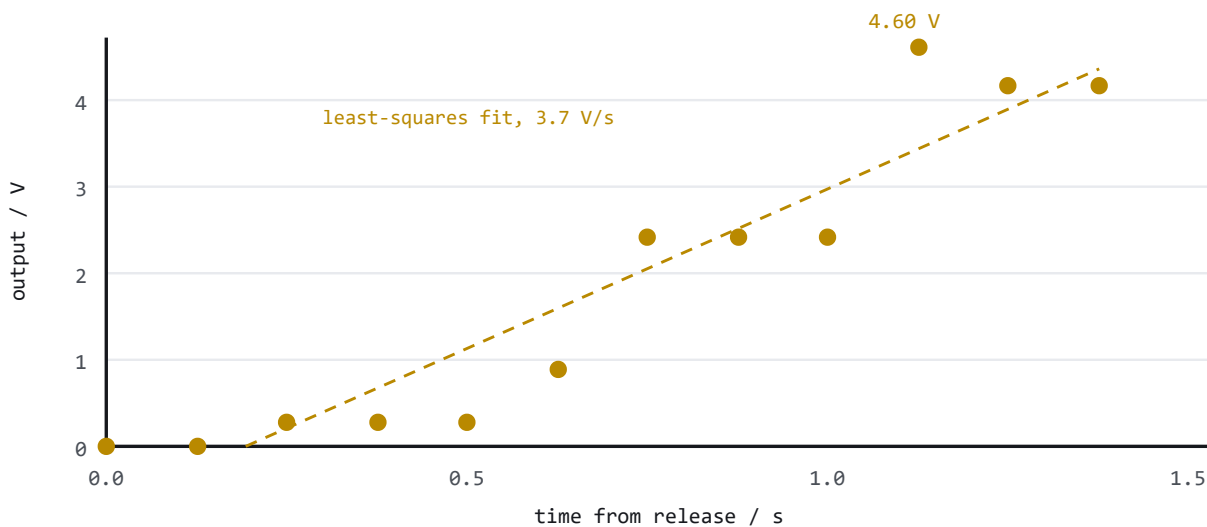


Figure 3 — Measured output during the drop, read off the meter frame by frame at 8 fps. Readings: 0.00, 0.00, 0.28, 0.28, 0.28, 0.88, 2.42, 2.42, 2.42, 4.60, 4.17, 4.17 V. The staircase is the meter's own update rate (roughly 3 Hz), not the machine. The dashed line is a least-squares fit to the twelve points: slope 3.7 V/s, intercept -0.7 V — that negative intercept is the meter's settling lag plus the slack taken up before the string began to pull.

Two conclusions come straight off that plot, and neither needed any assumption about the machine's construction.

The voltage rises linearly with time. Since $\varepsilon = k \omega_m$ and k is a constant of the build, a straight line in voltage is a straight line in shaft speed: the rotor is under a very nearly constant angular acceleration for the whole 1.4 s recorded. That is precisely what (16) plus a dominant inertia predicts, and it means the electrical braking of (12) was still small compared with the applied torque — the machine never reached its terminal speed during the drop. The falling mass was, in the strict sense, still accelerating when it hit the floor.

The gradient gives the machine constant, once the speed is known. The fitted slope is 3.7 V/s. The spool is 25 mm in diameter MEASURED, so the string leaves it at $r_s = 12.5$ mm, and the mass covers the drop in a few seconds, putting it near 0.5 m/s EST. at the point of the last reading — so $\omega_m \approx v/r_s \approx 40$ rad/s, about 380 rpm. Hence

$$k_{\text{meas}} = \frac{\varepsilon}{\omega_m} \approx \frac{4.2}{40} \approx 0.10 \text{ V}\cdot\text{s}/\text{rad} \quad (\text{vs. } k_{\text{ideal}} \approx 0.11 \text{ from 6b}) \quad (17)$$

The string wraps in a layer of finite thickness, so the effective radius grows a little as the spool fills and shrinks as it pays out; at 12.5 mm of radius a millimetre of winding is an 8% error on ω , which is small against the uncertainty in the fall speed itself.

THE RESULT

0.10 measured against 0.11 predicted. Those two numbers were arrived at along completely separate routes — one from magnet dimensions, turn count and winding geometry, the other from a falling mass, a spool radius and a voltmeter — and they agree. A machine assembled out of plumbing pipe and hot glue by someone who had not yet met calculus behaved the way the textbook says it should.

How much that agreement is worth

Less than it looks, and it would be dishonest to bank it. The dominant uncertainty is the fall speed: $\pm 30\%$ on v is $\pm 30\%$ on ω and therefore on k_{meas} , so the honest measured range is **0.08–0.14 V·s/rad**. The prediction has its own slack — chiefly B_{gap} , which I have taken as 0.15 T for neodymium blocks either side of a coil a few millimetres thick, and which could defensibly be argued anywhere between 0.10 and 0.20 T. What the comparison genuinely establishes is that the two are the same size — that no large effect is missing from the model — not that the model is accurate to 10%.

Three real effects sit inside that slack, all pushing the machine below its prediction:

- **Not every turn links the full flux.** Equation (6a) credits all 400 turns with the flux through the mean turn. The innermost turns enclose less area than that, the outermost enclose more but sit further from the pole; the mean is a fair approximation, not an exact one.
- **Alignment and phase.** Four coils in series only add arithmetically if their EMFs are in phase, which requires the coil spacing to match the pole spacing accurately. Coils glued into a printed holder by a fifteen-year-old are not perfectly indexed, and any phase error subtracts.
- **The meter is not an oscilloscope.** A cheap DMM after a bridge and a smoothing capacitor reads somewhere between the average and the peak of a rectified waveform. If it was reading closer to the average, the true peak EMF was nearer $4.2/0.64 \approx 6.6$ V, and the honest statement is that peak EMF lay somewhere in 4.2–6.6 V.

That last point resolves what would otherwise be a contradiction. Equation (10) says the blue LEDs need 4.4 V to light at all — about 44 rad/s, or 420 rpm, by equation (11) — and the meter never showed more than 4.60 V, yet the LEDs are clearly lit in the footage well before the end of the drop. A rectified waveform whose *peak* is around

6.6 V while its average reads 4.2 V explains both observations at once, and it is also why the crank and its 4:1 gearbox — not the falling mass — were the reliable way to light the thing.

Efficiency, to the nearest order of magnitude

Three blue LEDs drawing on the order of 5 mA at 3 V for the few seconds of the drop amount to roughly 0.2 J out of the 1.96 J that went in — an efficiency in the region of **10%** EST.. The lost 90% is not mysterious, and can be named in order of size:

- **Kinetic energy never converted.** The rotor is still spinning when the mass lands. Everything left in $I\omega^2/2$ at that instant never became light. Against a slowly-accelerating rotor this is easily the largest single term.
- **Rectifier drops.** Two diodes at ~0.7 V in series with a ~3 V load is roughly a third of the delivered voltage burned as heat before the LEDs see it.
- **Copper loss.** 31 Ω of source resistance against an LED load that is not remotely matched to it — equation (9) says this transfer is far off its optimum.
- **Bearing friction and string losses.** Small, but not zero, and the only ones I could actually feel.

9 Post-mortem: why the first prototype imploded

The first version, six months earlier, was cut from expanded polystyrene with a craft knife. Two magnet disks faced each other across a gap, exactly as in Figure 1, and after a handful of runs it destroyed itself: the disks pulled together and the webs between the magnet pockets folded inward. It never lit anything. I understood at the time that the magnets had done it. I did not understand that it was predictable with a single equation.

Two magnetised surfaces facing each other across a small gap attract with a force given by the Maxwell stress on the pole face:

$$F = \frac{B^2 A}{2\mu_0}, \quad \mu_0 = 4\pi \times 10^{-7} \text{ H/m} \tag{18}$$

The *square* in the numerator is the whole story. Per opposed pair of 20 × 10 mm magnets, at a pole-face field of 0.3 T EST.:

$$F_{\text{pair}} = \frac{(0.3)^2 \times 2 \times 10^{-4}}{2 \times 4\pi \times 10^{-7}} \approx 7.2 \text{ N} \Rightarrow F_{\text{total}} \approx 8 \times 7.2 \approx 57 \text{ N} \approx 5.8 \text{ kgf} \tag{19}$$

POLE-FACE FIELD	FORCE PER PAIR	EIGHT PAIRS	IN FAMILIAR TERMS
0.25 T	5.0 N	40 N	4 kg hanging off the disks
0.30 T	7.2 N	57 N	6 kg
0.35 T	9.7 N	78 N	8 kg

Six kilograms of axial pull, applied to a disk of expanded polystyrene about 100 mm across with eight rectangular holes cut through it. Two separate things then go wrong, and the second is the one that makes the failure sudden rather than gradual.

The material was asked the wrong question

Spread over the magnet faces, 7.2 N on 200 mm² is a bearing stress of about 36 kPa — comfortably under the ~100–150 kPa compressive strength of EPS foam, so a naive check passes. But the disk was never in pure

compression. The load arrives at eight discrete pockets near the rim and has to travel through the slender webs between them to reach the hub. Those webs are loaded in *bending*, and a foam plate with eight holes punched around its circumference has almost no bending stiffness left where it matters. Failure came at a fraction of the stress a compression check would have predicted.

The instability: the force grows as the gap closes

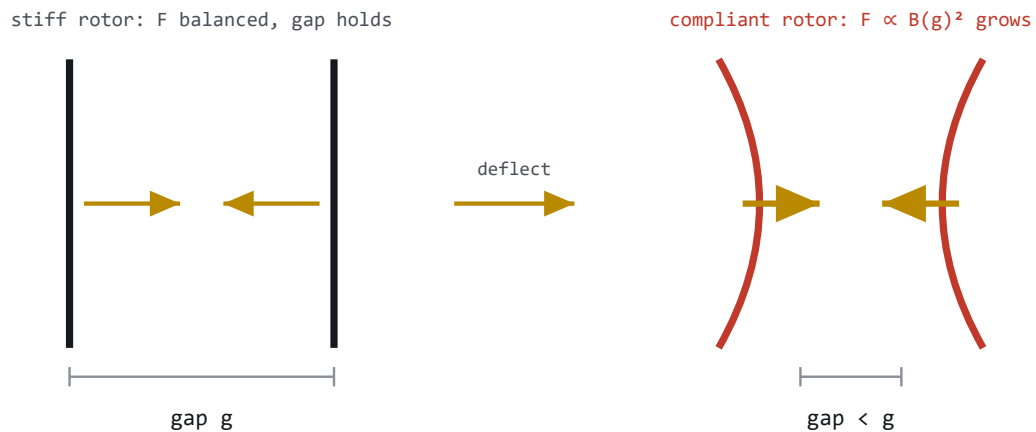


Figure 4 — Positive feedback in a compliant rotor. The disks deflect toward each other; the smaller gap raises the field, which raises the force by its square, which deflects them further. Past a critical stiffness there is no equilibrium to settle at, and the structure snaps through rather than sagging.

Field strength between the disks rises steeply as the gap closes, and force goes as the square of the field. So a small elastic deflection increases the load that caused it. Writing the structure as a linear spring of stiffness k_s against a load that grows as the gap shrinks, equilibrium requires

$$k_s > \frac{dF}{dg} \quad \text{at the operating gap} \quad (20)$$

Below that stiffness there is no stable gap: the rotor closes on itself. This is the same snap-through instability that makes electrostatic MEMS actuators "pull in", and it is why the polystyrene version did not sag slowly and then get fixed — it went from working badly to destroyed in one attempt.

WHAT THE FIX ACTUALLY WAS

Version two changed the material and nothing else about the concept: PLA at roughly a thousand times the elastic modulus of EPS foam, printed disks with the magnets in closed pockets, a copper-pipe shaft carried in real ball bearings, and a PVC frame that held both rotors at a fixed spacing whatever the magnets wanted. The 57 N did not go away — it was simply given a load path stiff enough that condition (20) was never in doubt. That is the whole difference between the two builds, and it is a materials-selection problem, not a magnetics one. Six months of frustration, one equation.

10 What I would measure if I built it again

The largest uncertainty in this document is speed. The spool radius is known, but the fall velocity behind equation (17) is still counted off video frames by eye, and ω is directly proportional to it — so k_{meas} is only as good as a hand-timed drop. In order of how much each would sharpen the analysis:

- **An actual tachometer.** An IR reflective sensor and one white mark on the rotor turns every voltage reading into a point on a $k = \varepsilon/\omega$ line, and the slope of that line is the machine constant with no geometry assumed at all. I built a mount for exactly this and never finished the instrumentation.
- **A known load resistor instead of LEDs.** A resistor of known value makes power directly computable as V^2/R , and sweeping R_L maps out equation (9) experimentally — the maximum-power point is a thing you can see on a bench.
- **An oscilloscope across one coil, before the bridge.** This removes the meter ambiguity of §8 permanently: peak, RMS, waveform shape and electrical frequency all become directly readable, and f gives the shaft speed for free through equation (4) — a tachometer built out of the generator itself.
- **Coast-down testing.** Spin the flywheel up, disconnect the load, record the decay: that separates bearing and windage losses from electrical ones, which is the only way to close the energy budget of §8 honestly.

A Symbols, values and where they came from

SYMBOL	QUANTITY	VALUE USED	PROVENANCE
N	Turns per coil	≈ 400	from the winding window, eq. (8a)
—	Winding window	$5 \times 6 \text{ mm}$	the printed holder MEASURED
d_{ins}	Wire OD over enamel	0.25 mm	builder's record MEASURED
D_{mean}	Mean turn diameter	12 mm	6 mm former + 6 mm build
—	Coils, in series	4	photographs MEASURED
—	Wire gauge	AWG 32 (0.203 mm)	builder's record MEASURED
p	Pole pairs	4 (8 magnets/disk)	counted from photographs MEASURED
A_m	Magnet pole face	$20 \times 10 \text{ mm}$	scaled from photographs EST.
B_{gap}	Working-gap field	$\approx 0.15 \text{ T}$	neodymium blocks across the coil EST.
B_{face}	Pole-face field	$0.25 - 0.35 \text{ T}$	typical N35 block EST.
R_{coil}	Coil resistance	$\approx 7.8 \Omega$ each	computed from (7)–(8b)
m	Flywheel plate	2 kg (5 kg tested)	marked on the plate MEASURED
r_o, r_i	Plate radii	90 mm, 12.5 mm	standard plate geometry EST.
I	Flywheel inertia	$8.3 \times 10^{-3} \text{ kg} \cdot \text{m}^2$	computed from (13)
G	Gearbox ratio	4:1	CAD model MEASURED
m_ϕ, h	Drop mass and height	200 g, 1.00 m	the experiment MEASURED
r_s	Spool radius	12.5 mm ($\emptyset 25 \text{ mm}$)	builder's record MEASURED
v	Fall speed at last reading	$\approx 0.5 \text{ m/s}$	counted off video frames EST.
ε	Peak output, drop test	4.60 V (meter)	read frame by frame MEASURED
V_f	Blue LED / diode drop	3.0 V / 0.7 V	standard device data
ρ_{Cu}	Copper resistivity	$1.68 \times 10^{-8} \Omega \cdot \text{m}$	reference value
μ_0	Permeability of free space	$4\pi \times 10^{-7} \text{ H/m}$	defined constant

Every number in this document that is not marked MEASURED is an estimate, and the estimates are deliberately conservative rather than flattering. The machine still exists, on a shelf in São Paulo; if it is ever put back on a bench with a tachometer and a load resistor, §8 and §10 are the pages to redo first.